

جامعة الأميرة
نورة بنت عبد الرحمن
كلية علوم الحاسب والمعلومات



Princess Nourah bint
Abdulrahman University
College of Computer and
Information Sciences

CS 385: Software Engineering Term Project

Title: Surea Delivery Drones

Contents

1. Chapter 1	3
1. System Description.....	3
2. Development Process Model.....	3
3. System Requirements	3
3.1 User Functional Requirements:	3
3.2 System Functional Requirements (Multilevel).....	4
4. Non-Functional Requirements (NFRs)	4
2. Chapter 2.....	5
Context Diagram.....	5
System Use-case Diagram with description	6
Use case scenario (MSS +Extensions) [two per student].....	7
Use case 1Click or tap here to enter text.....	7
Use case 2 Click or tap here to enter text.....	8
Use case 3 Click or tap here to enter text.....	9
Use case 4 Click or tap here to enter text.....	10
Use case 5 Click or tap here to enter text.....	Error! Bookmark not defined.
Use case 6 Click or tap here to enter text.....	Error! Bookmark not defined.
Use case 7 Click or tap here to enter text.....	10
Use case 8 Click or tap here to enter text.....	Error! Bookmark not defined.
3. Chapter 3	15
System Class Diagram	15
Sequence Diagrams	16
Sequence diagram for Click or tap here to enter text.	16
Sequence diagram for Click or tap here to enter text.	17
Sequence diagram for Click or tap here to enter text.	18
Sequence diagram for Click or tap here to enter text.	19
4. Chapter 4	20
Activity Diagram [one per student].....	20
Activity Diagram for Click or tap here to enter text.....	20
Activity Diagram for Click or tap here to enter text.....	21
Activity Diagram for Click or tap here to enter text.....	21
Activity Diagram for Click or tap here to enter text.....	22
State Diagram	22
State Diagram for Click or tap here to enter text.	22
5. Chapter 5.....	23

Architecture Diagram.....	23
6. Chapter 6.....	24
Reference.....	24

Phase 1 – Surea Delivery Drones

1. Chapter 1

1. System Description

Surea Delivery Drones is a system that manages a fleet of drones to deliver parcels locally. It automates order scheduling and drone assignment, optimizes routes for safety and battery limits, and monitors deliveries in real time. Customers receive delivery status updates and proof-of-delivery via OTP or QR codes. The system also provides dashboards for admins and maintenance notifications for drone technicians, using simulated telemetry and mock external services for maps, weather, and notifications.

2. Development Process Model

The **Waterfall model** is used for this project. It is suitable because the project has clearly defined requirements, a structured workflow (order creation → drone assignment → delivery → proof-of-delivery), and deliverables that can be planned in sequential phases. Each phase (requirements, design, implementation, testing) can be completed before moving to the next, ensuring stability for this critical delivery system.

3. System Requirements

3.1 User Functional Requirements:

- 1.The system shall allow a customer to register and create a delivery order with pickup and delivery addresses and parcel details.
- 2.The system shall provide an estimated price for the delivery based on distance and parcel weight.
- 3.The system shall allow the customer to track their parcel in real time and view live drone movement on a map.
- 4.The system shall send notifications to the customer when the delivery status changes to In Progress, Delivered, or Failed.
5. The system shall provide a user-friendly FAQ section to assist customers with common inquiries.
6. The system shall enable customers to rate their delivery experience and provide feedback.
7. The system shall allow administrators to view all active drones, their battery levels, and assigned deliveries on a centralized dashboard.
8. The system shall enable drone technicians to receive automatic maintenance alerts when a drone’s battery health or flight stability drops below safe thresholds.
9. The system shall allow customers to choose between standard and express drone delivery options

10. The system shall automatically notify customers when weather conditions delay delivery

3.2 System Functional Requirements (Multilevel)

1. Customer Order Creation

- 1.1 The system shall allow input of pickup and delivery addresses, parcel type, and customer details.
- 1.2 The system shall validate addresses and required fields before confirming the order.

2. Real-Time Tracking

- 2.1 The system shall provide a live map interface showing the current location of the drone.
- 2.2 The system shall update the drone's location on the map every 30 seconds.

3. Notifications

- 3.1 The system shall allow customers to opt-in for notifications via SMS, email, or app alerts.
- 3.2 The system shall notify customers of any status changes (e.g., In Progress, Delivered, Failed).

4. Drone Monitoring

- 4.1 The system shall display all active drones on an admin dashboard with real-time telemetry data such as altitude, battery percentage, and signal strength.
- 4.2 The system shall highlight drones with low battery or signal loss by changing their status indicator color to red.

5. Weather Delay Notification

- 5.1 The system shall integrate with a live weather API to detect severe or unsafe flying conditions along the drone's route.
- 5.2 The system shall automatically send real-time notifications to customers and admins via SMS and email if a delivery is delayed, rescheduled, or canceled due to weather conditions.

4. Non-Functional Requirements (NFRs)

1. Performance

- **Goal:** The system shall update shipment tracking locations in near real time.
- **Metric:** 95% of tracking API responses shall occur within ≤ 2 seconds.

2. Reliability

- **Goal:** The system shall operate continuously during normal use without failure.
- **Metric:** The system shall maintain at least 95 % uptime during 50 concurrent simulated deliveries.

3. Security

- **Goal:** The system shall protect user data and transaction information.
- **Metric:** The system shall implement encryption for all sensitive data in transit and at rest, achieving 100% compliance with industry standards.

4. Usability

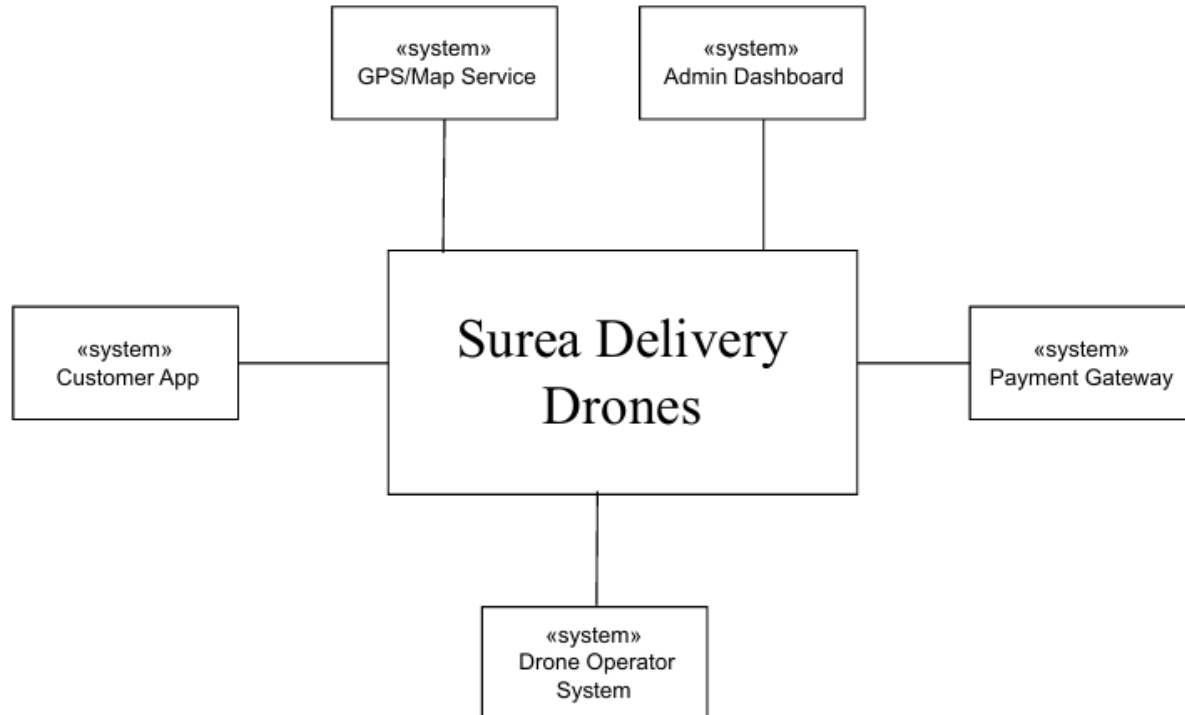
- **Goal:** The admin and technician dashboards shall be easy to use and navigate.
- **Metric:** At least 90% of users shall successfully complete basic tasks (e.g., viewing drone status or acknowledging alerts) during usability testing.

5. Scalability

The system shall support growth in both users and drones without any performance degradation, and shall handle up to 10 000 concurrent user requests while maintaining an average response time of 3 seconds or less.

2. Chapter 2

Context Diagram

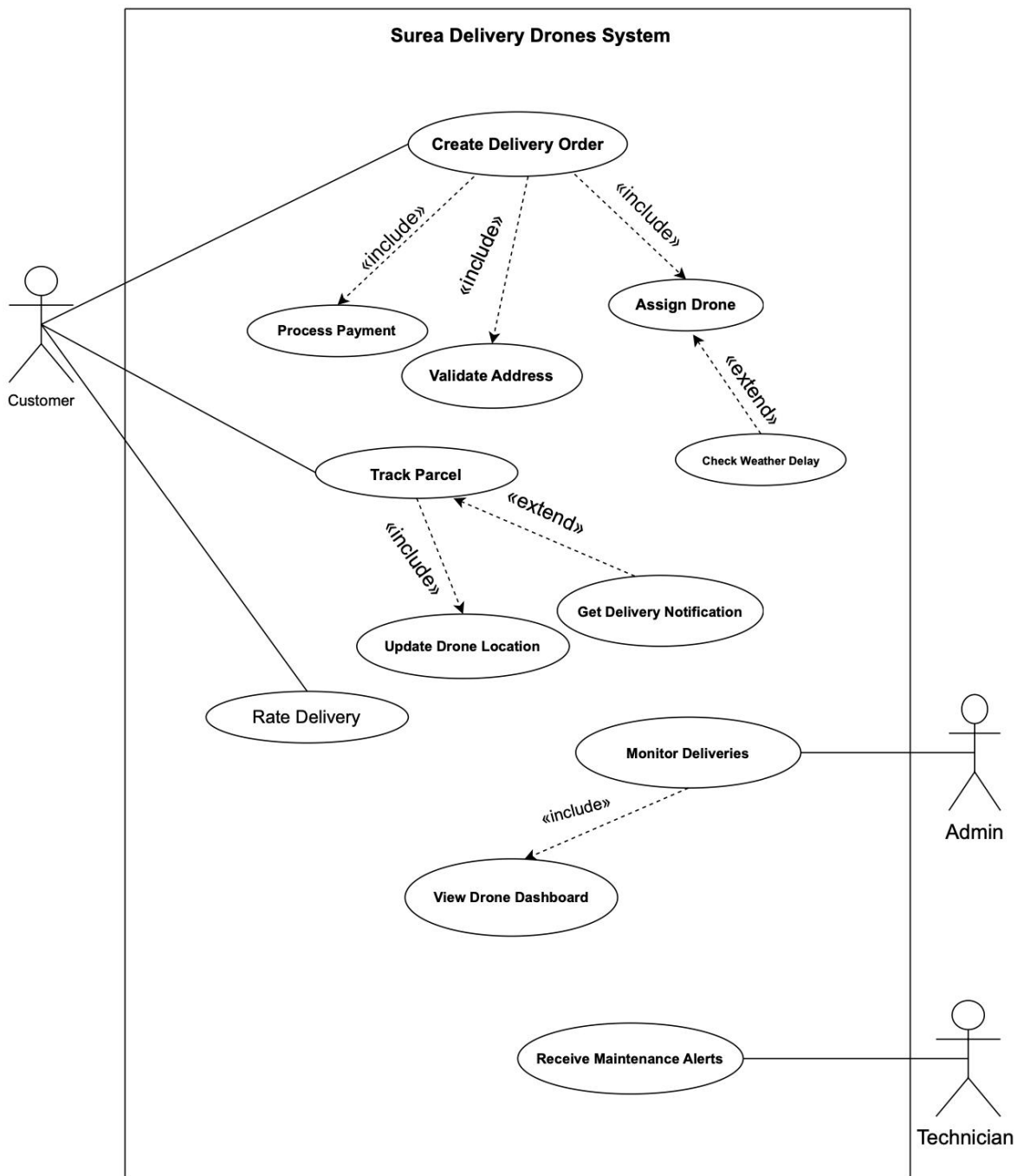


System Use-case Diagram with description

When a **customer** creates a delivery order, the system must process the payment, validate the address, and assign a drone as part of the main process.

During delivery, the parcel is tracked, and the drone's location is regularly updated for real-time monitoring. If special conditions occur, such as bad weather, the system may check for weather delays, or send delivery notifications to keep the customer informed.

After the delivery, the customer can rate the service, while the **admin** monitors deliveries through the drone dashboard and the **technician** receives maintenance alerts when needed.



Use case scenario (MSS +Extensions)

Use case 1 [Create Delivery Order]

Use-case Name	Create Delivery Order	
Goal	To allow the customer to create a new drone delivery request by entering all required parcel and address details, confirming payment, and receiving an assigned drone with tracking information.	
Actors	Customer	
Main Success Scenario	1	The Customer opens the Surea Delivery Drones application.
	2	The system displays the “Create Delivery Order” page.
	3	The Customer enters the pickup and delivery addresses, parcel type, and weight.
	4	The system validates the entered details and calculates an estimated delivery cost.
	5	The Customer confirms the order and proceeds to payment.
	6	The system processes the payment and confirms the order.
	7	The system assigns an available drone and displays the order summary with tracking information
Extension	1	If required fields are left empty, the system displays an error message requesting completion.
	2	If the address is invalid, the system prompts the Customer to re-enter a valid one
	3	If payment fails, the system notifies the Customer and allows retry or alternative payment methods.
	4	If no drone is available, the system places the order on hold and informs the Customer that delivery will start once a drone becomes available.

Use case 2 [Assign Drone]

Use-case Name	Assign Drone	
Goal	Automatically assign the most suitable drone to a confirmed delivery order and provide ETA to the customer	
Actors	Customer	
Main Success Scenario	1	System detects a newly confirmed delivery order.
	2	System checks available drones and their current locations.
	3	System validates constraints (payload, battery level, weather, route distance).
	4	System selects the most suitable drone based on availability and constraints.
	5	System reserves and assigns the drone to the delivery.
	6	System updates the order status and drone status (e.g., “Reserved/Dispatched”).
	7	System notifies the Customer with assigned drone details and ETA.
Extension	1	If no drones are available, the system places the order in a queue and informs the customer about the delay.
	2	If delivery limits like weight or weather aren’t met, the system reassigns another drone or reschedules with a new arrival time.
	3	If the assigned drone’s battery is too low, the system selects a different one.
	4	If the notification fails, the system resends it and records the failure

Use case 3 [Process Payment]

Use-case Name	Process Payment	
Goal	To let the customer complete the order payment securely and confirm the transaction through the payment gateway.	
Actors	Customer	
Main Success Scenario	1	The customer reviews the delivery order and total price
	2	The system displays available payment options
	3	The customer selects a method and enters payment details
	4	The system sends the payment request to the payment gateway
	5	The payment gateway verifies and approves the transaction
	6	The system receives confirmation and marks the order as paid
	7	The customer receives a payment receipt and order confirmation
Extension	1	If payment details are missing or invalid, the system asks the customer to correct them.
	2	If the payment gateway times out, the system retries and notifies the customer to wait
	3	If OTP or verification fails, the system cancels the attempt and allows a retry or another method.
	4	If the payment is declined, the system alerts the customer and keeps the order pending
	5	If duplicate payment attempts occur, the system ensures only one transaction is processed

Use case 4 [Receive Maintenance Alerts]

Use-case Name	Receive Maintenance Alerts	
Goal	Notify the Technician about any maintenance issues or abnormal drone conditions to ensure system safety.	
Actors	Technician	
Main Success Scenario	1	The system continuously monitors drone telemetry for performance metrics such as battery health, motor stability, and flight time
	2	If the battery or stability of a drone falls below threshold values, the system will automatically create a maintenance alert.
	3	A notification will appear on the Technician's Maintenance Dashboard
	4	The Technician views the alert details: drone ID, issue type, timestamp, and status.
	5	The Technician acknowledges the alert and schedules a maintenance check.
	6	The system logs the maintenance activity for recordkeeping.
	7	The Technician performs the required maintenance, and the system updates the drone's status to "Operational."
Extension	1	On failure to send the notification, the system tries sending it again and logs the failure
	2	If multiple drones report simultaneously, warnings are provided according to the level of seriousness, such as critical battery failure first.
	3	If the Technician does not acknowledge the alert in the stipulated time, the system escalates the notification to the Admin
	4	If maintenance data fails to save, the system retries and notifies the technician to confirm manually.

Use case 5 [Track Parcel]

Use-case Name	Track Parcel	
Goal	The goal of this use case is for the customer to track the status of their parcel throughout the delivery process.	
Actors	Customer	
Main Success Scenario	1	The customer accesses the delivery system.
	2	The customer selects the 'Track Parcel' option.
	3	The system prompts the customer to enter the parcel tracking ID.
	4	The customer inputs the tracking ID and submits the request.
	5	The system retrieves the parcel information from the database.
	6	The system displays the current status and location of the parcel to the customer.
	7	The customer reviews the parcel information and takes any necessary actions (e.g., contact support if there are issues).
Extension		If the customer enters an invalid order ID, the system displays an error message indicating that the parcel cannot be found
		The customer can contact customer service directly from the tracking page if they have any concerns.

Use case 6[Rate Delivery]

Use-case Name	Rate Delivery	
Goal	To allow the customer to provide feedback on the delivery service, enhancing overall service quality.	
Actors	Customer	
Main Success Scenario	1	The customer receives a notification that their delivery has been completed.
	2	The customer accesses the delivery history through the application or website.
	3	The customer selects the delivery they wish to rate.
	4	The customer provides a rating (typically on a scale of 1 to 5 stars) and may leave optional comments.
Extension		If the customer attempts to rate a delivery that does not exist in their history, the system displays an error message indicating that the delivery cannot be found.
		If the customer chooses not to leave a comment, they can submit the rating with just the star selection.

Use case 7[Monitor Deliveries]

Use-case Name	Monitor Deliveries	
Goal	To enable the admin to oversee and manage ongoing drone deliveries, ensuring efficient operations and timely customer service.	
Actors	Admin	
Main Success Scenario	1	The admin logs into the drone management dashboard.
	2	The admin reviews the status of each delivery in real-time.
	3	If any delivery is delayed or flagged for issues (e.g., bad weather, technical problems), the system alerts the admin.
	4	The admin can take necessary actions, such as notifying the customer about delays or rerouting the drone if needed.
	5	The admin can access historical delivery data to identify trends or issues.
	6	The admin generates reports on delivery performance for analysis and future improvements.
Extension		If the admin encounters a system error while accessing the dashboard, an error message is displayed, prompting the admin to refresh or contact support.
		If the admin identifies a recurring issue in delivery performance, they can escalate the matter to the technician for further investigation.

Use case 8 [View Drone Dashboard]

Use-case Name	View Drone Dashboard	
Goal	To allow the admin to monitor drone operations, including delivery status and maintenance needs, through a centralized dashboard.	
Actors	Admin	
Main Success Scenario	1	The admin logs into the drone management system.
	2	The admin navigates to the drone dashboard.
	3	The system displays the current status of all drones, including their locations and delivery statuses.
	4	The admin reviews the information on the dashboard.
	5	If a drone is delayed or requires maintenance, the system notifies the admin.
	6	The admin can initiate maintenance requests if needed.
	7	The admin logs out of the system.
Extension		If the admin encounters an issue with the dashboard, they can contact technical support for assistance.
		The admin can filter the displayed information based on specific parameters (e.g., delivery status, location).

3. Chapter 3

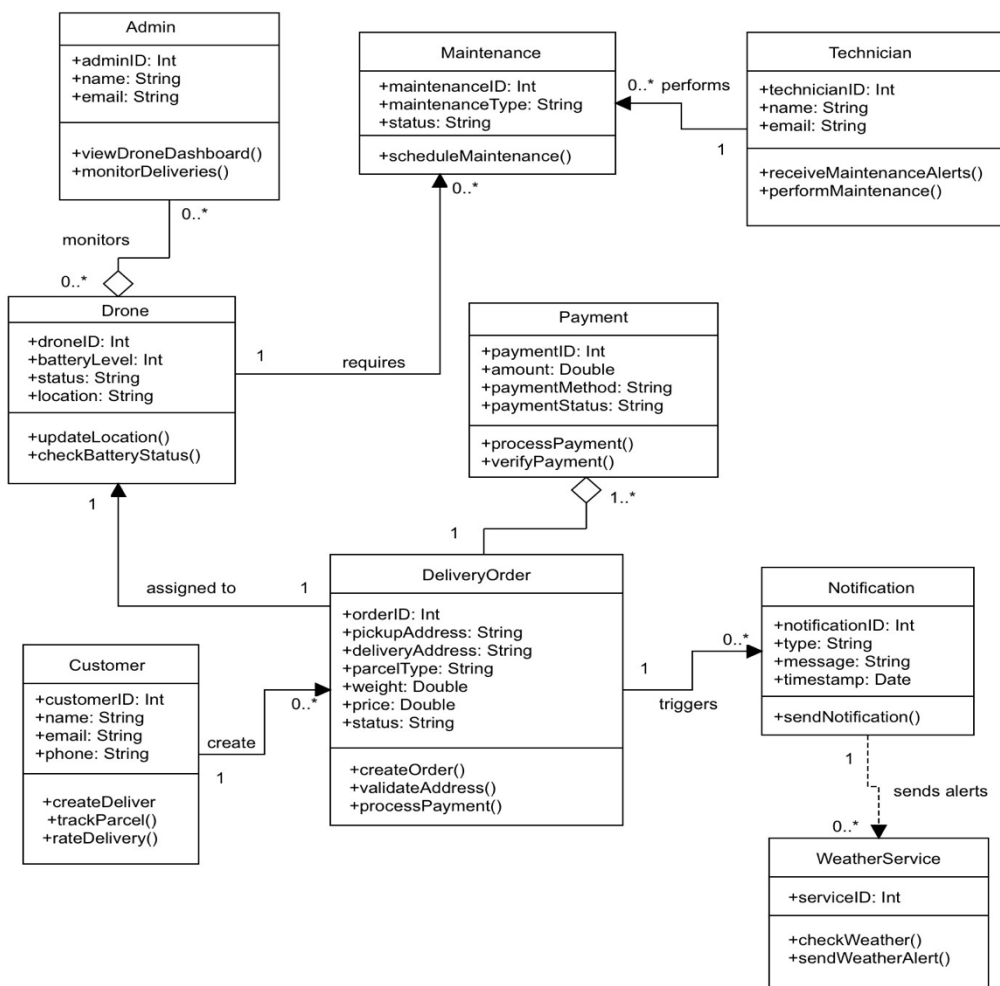
System Class Diagram

The class diagram for the Sura Delivery Drones System represents the main entities and their relationships within the delivery process.

It includes classes such as Customer, Admin, Technician, DeliveryOrder, Drone, Payment, Maintenance, Notification, and WeatherService.

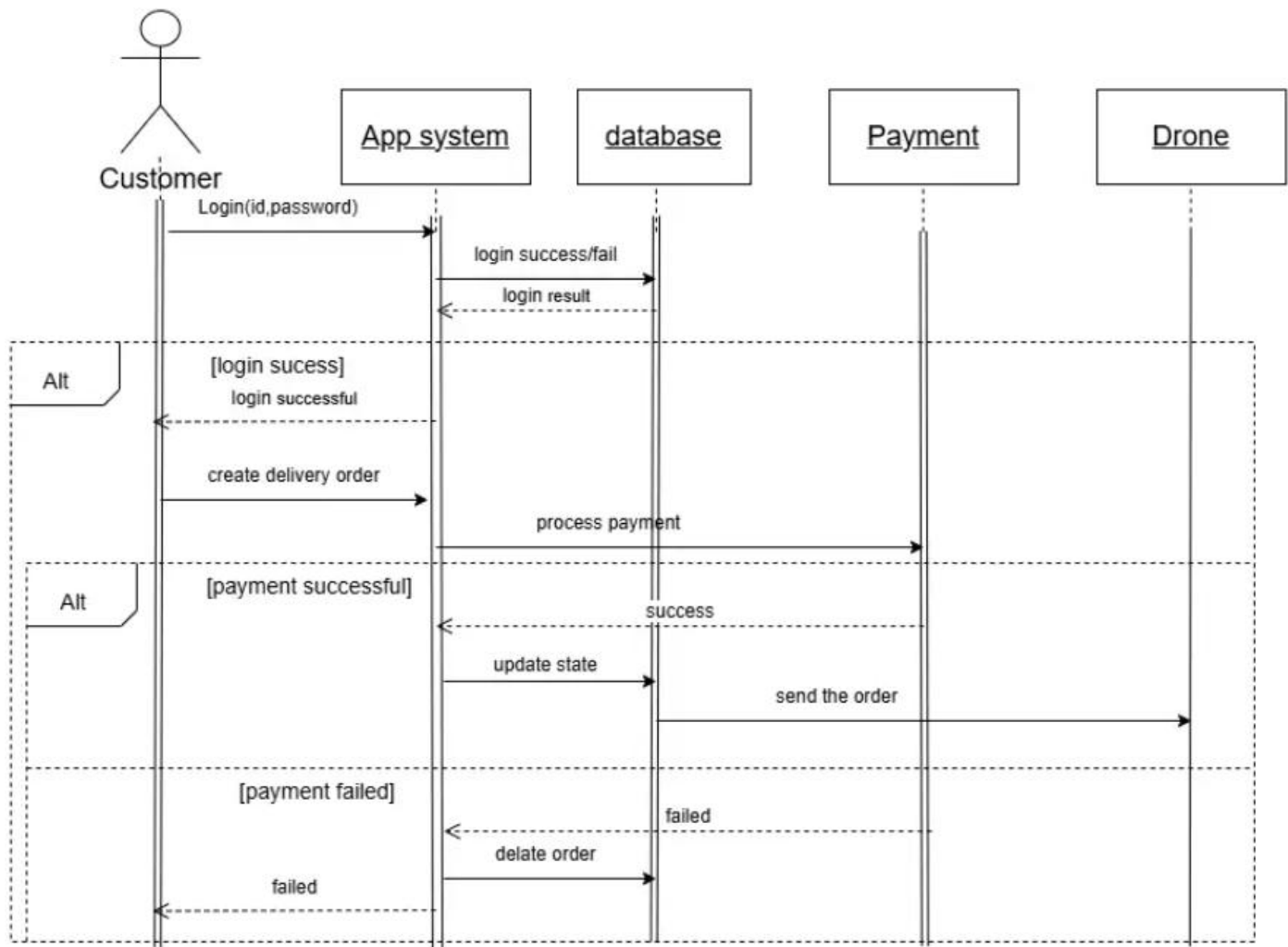
The Customer creates delivery orders, makes payments, and tracks parcels, while the Admin monitors drone activity and system performance. Technicians handle drone maintenance when alerts are received. DeliveryOrder acts as the core class, linking payments, assigned drones, and delivery notifications. Drone objects record status, battery level, and location, and connect to Maintenance records. WeatherService depends on Notification to send alerts in unsafe conditions.

The diagram shows associations (e.g., Customer–Order), aggregations (Admin–Drone, Drone–Maintenance), and dependencies (WeatherService–Notification), clearly defining how the system’s components interact to enable

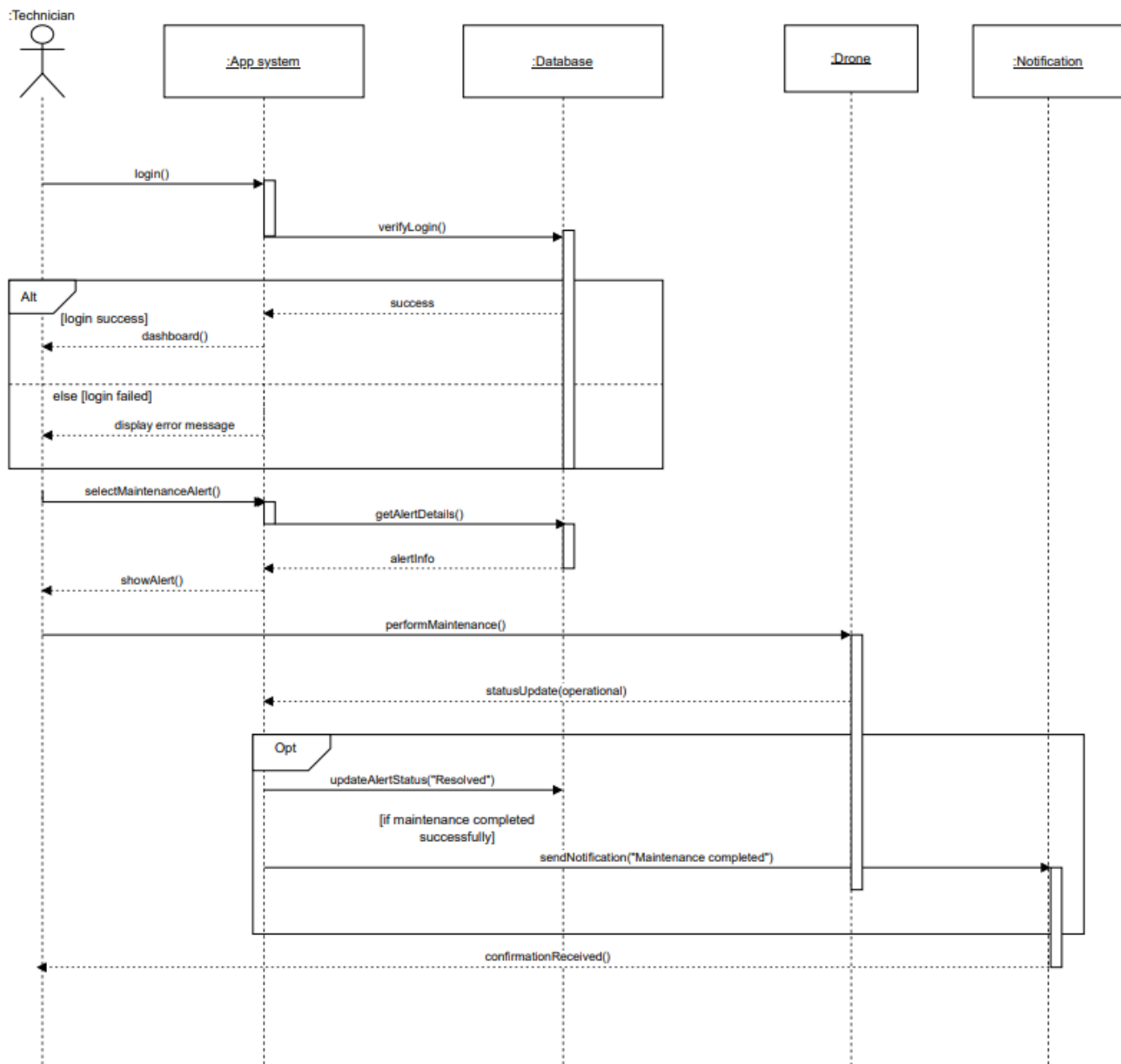


Sequence Diagrams

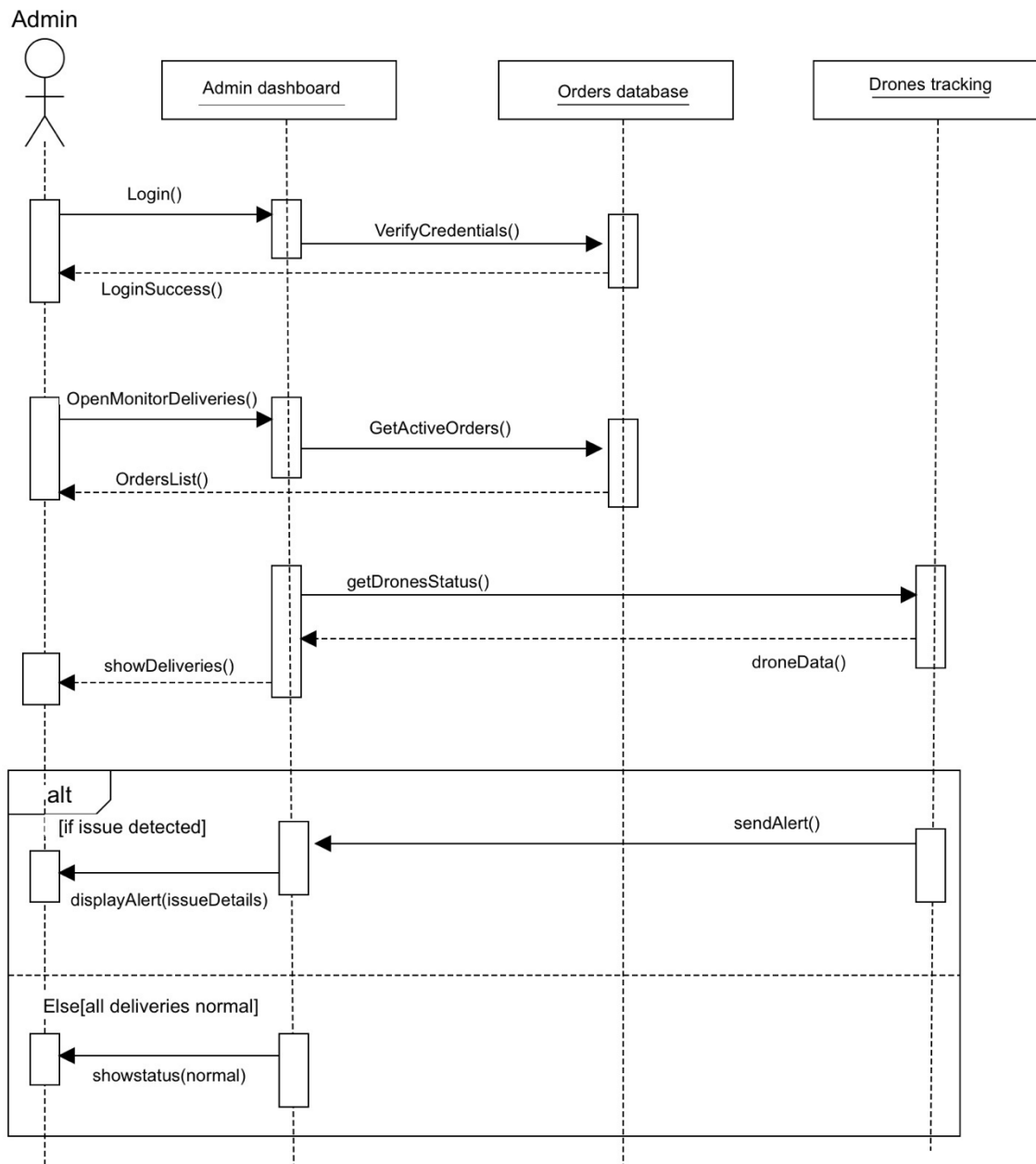
Sequence diagram for Login and create order



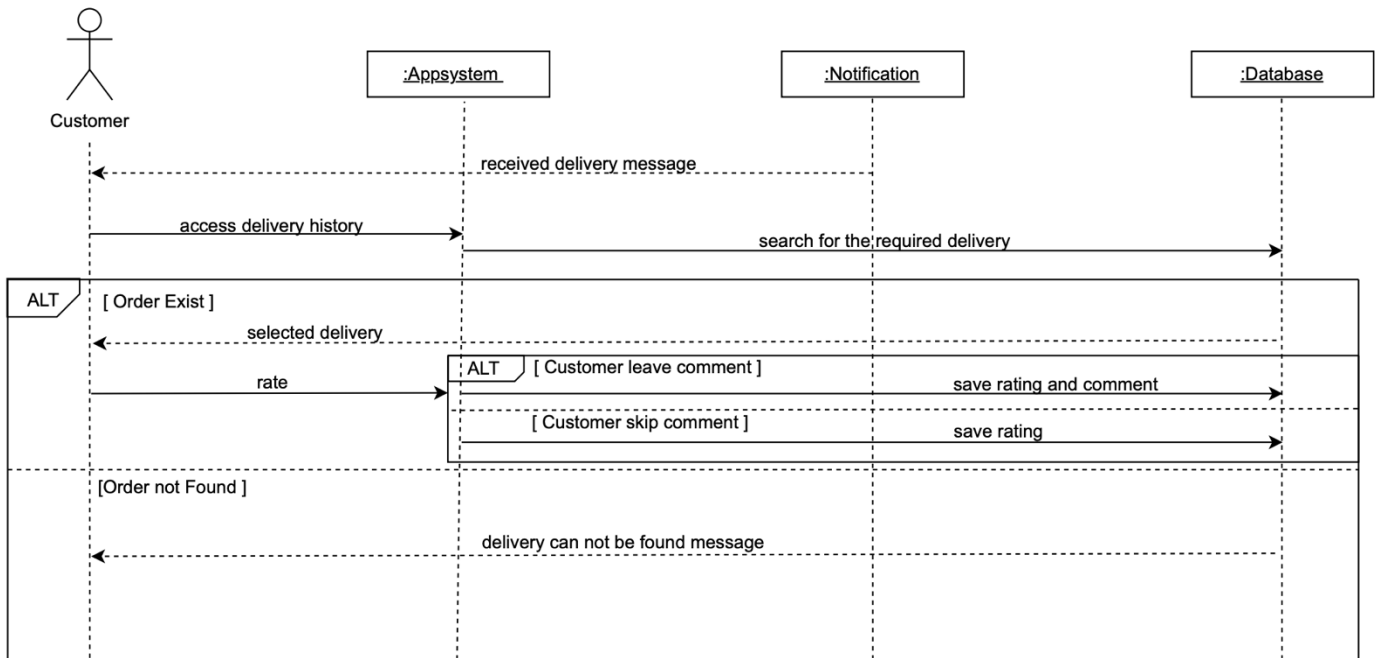
Sequence diagram for Receive maintenance alerts



Sequence diagram for admin monitoring drone delivery.



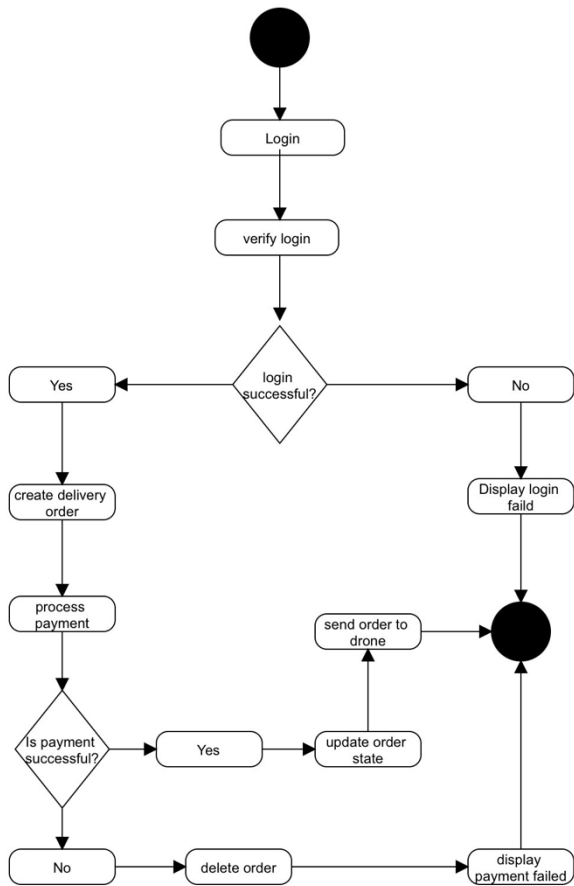
Sequence diagram for Rate Delivery



4. Chapter 4

Activity Diagram [one per student]

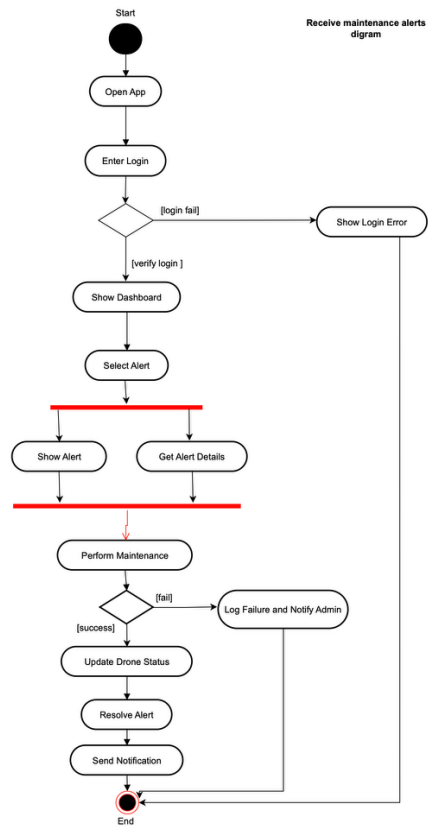
Activity Diagram for Login and create order



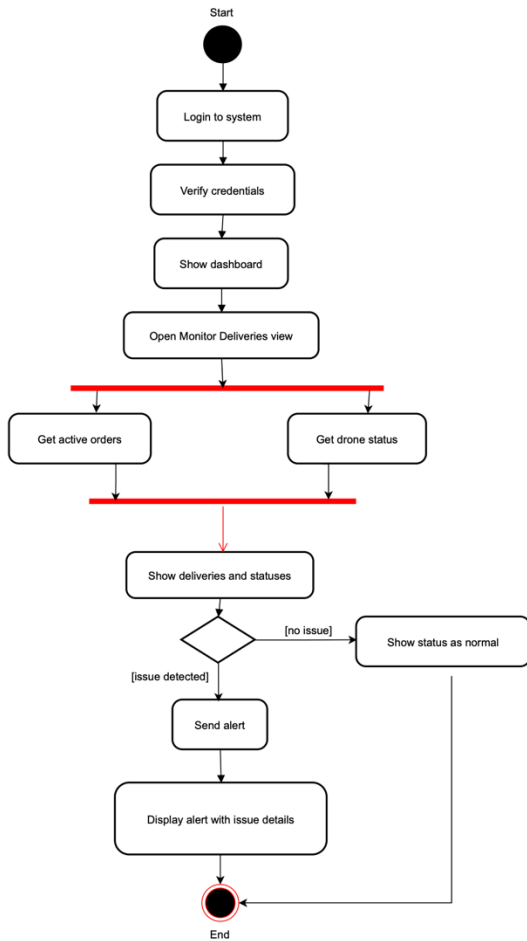
Activity Diagram for Rate Delivery



Activity Diagram for Receive maintenance alerts



Activity Diagram for C admin monitoring drone delivery.



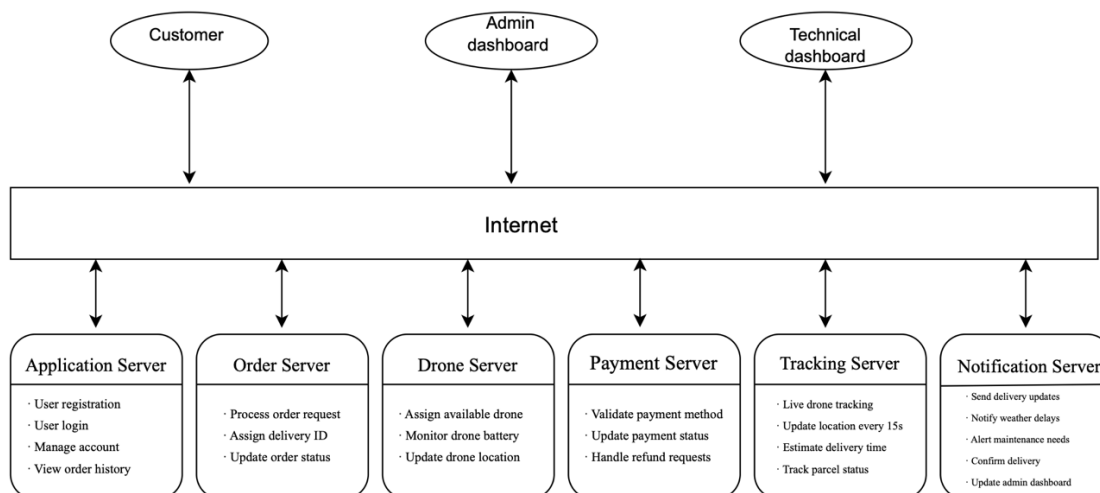
State Diagram

State Diagram for *Click or tap here to enter text.*

[Draw a state diagram for one object in your system (if exist)]

5. Chapter 5

Architecture Diagram



We chose this design mainly because it can grow easily. With a client-server system, we can add more servers when we get more users.

This way, the server can manage the extra work without problems. Another reason is that it helps us keep data organized. By using one main server, we can store user data safely. This makes sure the information is secure and easy to back up.

6. Chapter 6

Reference

1. Sommerville, I. Software Engineering, 10th Edition. Pearson Education.
2. Pressman, R. S., & Maxim, B. R. Software Engineering: A Practitioner's Approach. McGraw-Hill.
3. UML Activity Diagrams – UML.org
4. Waterfall Software Development Model – Tutorialspoint
5. Drone Delivery System Articles – IEEE Xplore
6. Class and Sequence Diagram Basics – Visual Paradigm Documentation